

NCTU-EE IC LAB – Fall 2018

Final Exam 2019/01/09 13:20~15:10

1. (a) Fig. 1 shows the basic 2 Flip-Flop synchronizer block diagram. Explain why a synchronizer is required in clock domain crossing (CDC) circuit. (5%)

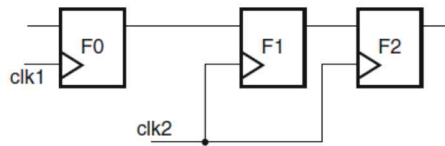


Fig. 1 2FF synchronizer

- (b) Fig. 2 shows the pulse synchronizer which we used in Lab07. Different from the synchronizer in Fig. 1 which can pass a value, the pulse synchronizer can pass a pulse. Try to complete the following timing diagrams to verify it can pass a pulse between two clock domains either when clk1 is faster than clk2 (5%) and when clk1 is slower than clk2 (5%).

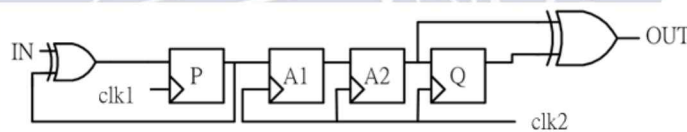
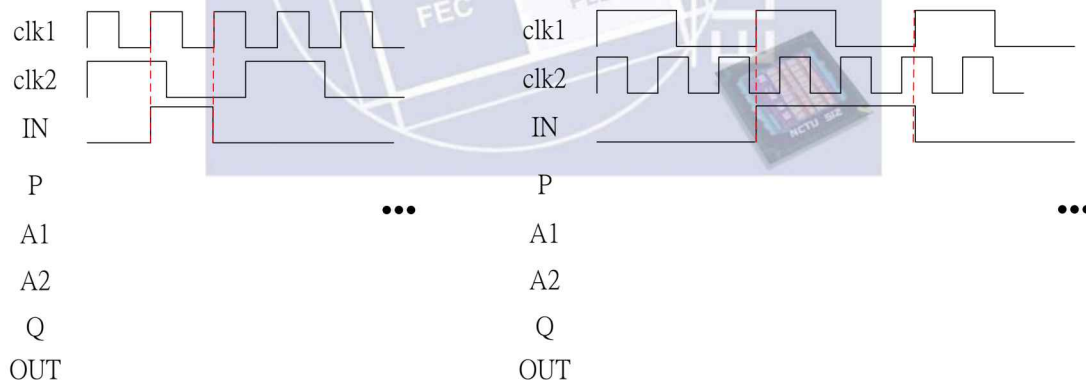


Fig. 2 pulse synchronizer



(Complete these two timing diagrams)

2. In Lab07, we inserted synchronizers between multi clock domains. However without further settings, the slack will be negative in synthesis and timing violation will occur in gate level simulation. Express what action have done to remove the negative slack issue in synthesis (5%) and timing violation issue in gate level simulation (5%).

3. Shift register is a common method to save sequential input signal without making too much area (as shown in the following Verilog code).

```
always@(posedge clk) begin
    q1 <= in;
end
always@(posedge clk) begin
    q2 <= q1;
end
always@(posedge clk) begin
    q3 <= q2;
end
...
always@(posedge clk) begin
    q25 <= q24;
end
```

However, it is not suitable for clock gating design since all flip-flops should work when input is coming. If the percentage of inputting signal time is small, it is ok since flip-flop can be shut off when input is not coming. However when inputting signals dominate the design working cycle, more power will be consumed. How will you try to make less power consumption for saving long-term input signal? (Simply describe your method and simply write down the Verilog code) (6%)

4. Fig. 3 shows the AND-based and OR-based clock gating cells:



Fig. 3 AND-based and OR-based clock gating cells

- (a) awake signal and sleep signal should be changed in a certain region (according to clock signals), or else the gated clock output will get glitches. Please specify the regions for both cells that awake / sleep should not change and use timing diagram to show that changing may result in glitches. (7%)

(Hint: you can let inputs: sleep / awake also have glitches that it will be easier to find the answer region)

- (b) Latch based clock gating cells as shown in Fig. 4 solve the glitch problems, which becomes the famous clock gating cells nowadays. Please use timing diagrams again that the glitches problem in 4.(a) can be fixed. (7%)

(Your timing diagram should contain node t in both cases)

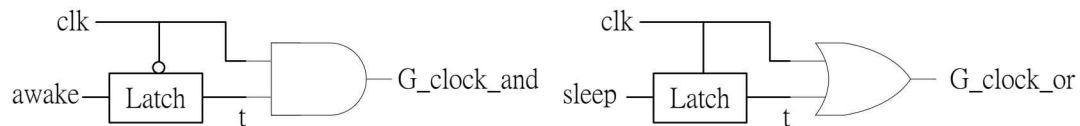


Fig. 4 Latch-based clock gating cells

5. Following is a covergroup declaration:

```
covergroup final @(posedge clk);
    option.per_instance = 1;
    option.at_least = 100;
    signal0: coverpoint inf.signal_0{
        bins signal_0 [] = ([0:4] => [0:4]);
    }
    signal1: coverpoint inf.signal_1{
        bins signal_1 = 1 ;
    }
    signal2: coverpoint inf.signal_2{
        option.auto_bin_max = 16;
    }
    signal3: coverpoint inf.signal_3{
        bins s3_0 = 0;
        bins s3_1 = 1;
        bins s3_2 = 2;
    }
    cross signal1 , signal3 ;
endgroup
```

(a) If signal_2 is a 10-bit signal, how many numbers will contain in one bin (in coverpoint signal2)? (3%)

(b) In what condition that this covergroup will achieve 100% coverage?
(You need to write down each condition clearly) (15%)

6. For using JasperGold in Lab10, you can choose to show the COI results and proof core results with the provided assertions, describe the difference between these two results. (5%)

7. In APR in Lab11. Different from .db files used in synthesis. We need to provide
- (a) .lib files (Hint: this one is the same as .db files, just with different format)
 - (b) .cdb files
 - (c) .lef files
 - (d) .captbl files and .tch files
 - (e) .gds files: These files contain the physical information like metals / polys / substrates of each cell. At the end of APR, we need to merge the current layout with the .gds files of all cells to generate the .gds file for whole layout.
- Please describe what these files (a)~(d) are and the purposes of providing these files. (12%)
8. Briefly describe what pad limited design and core limited design are in APR. (5%)
9. After we finished our layout with our APR tool, we have run the power rail analysis to see the IR drop result in Lab12:
- (a) Please specify what IR drop is? (5%)
 - (b) What is the effect of the IR drop? (at least 2) (5%)
 - (c) How to avoid the IR drop? (5%)

